

**Addressing the rise of
cyber-nuclear threats to
achieve global security**

Counter-Terrorism Committee (CTC)

I. Introduction

"We reject techniques like torture regardless of whether they're effective or ineffective because they are barbaric and harmful on a broad scale. [...] We should never be taking down power plants unless that is absolutely necessary to ensure our continued existence as a free people."

-Edward Snowden

Ever since the world entered the atomic age in 1945, humanity has found itself multiple times on the brink of nuclear war, with 8 other nations besides the United States of America acquiring the bomb.

Following the digital revolution of the 1990s, all nuclear powers of the world have, in some way, pursued the integration of new technologies into their arsenal; the mechanical is being phased out in favour of the digital, and jobs that required dozens of well-trained and conscious engineers and middlemen have been replaced by lines of code.

This seemingly natural change, however, has stirred up experts' concerns¹ regarding the safeguards of humanity in front of a nuclear end of the world. A simple computer virus has the potential to lead to a deadly international crisis.

II. Definition of Key Terms

- ***Launch spoofing***

Faking signals to trick a computer system. This is usually done to either trick a system into launching a nuclear strike without permission or to trigger false alarms by making early-warning sensors think a launch has happened when it hasn't.

- ***Worm***

Malware that can replicate itself and spread independently across computers and networks. Unlike traditional viruses, worms don't require human action to propagate.

- ***Extremist groups***

Organisations that promote radical ideologies, using violence for achieving political, religious and social goals.

- ***Command override***

Bypassing a system's automatic controls to take direct charge. This can be a legitimate safety feature used by a human operator to stop an accident, or it can be a cyberattack where a hacker forces the system to ignore its normal rules and obey their commands instead.

¹ <https://www.dw.com/en/biden-putin-summit-experts-warn-of-cyber-nuclear-nightmare/a-57906147>

- ***Silent Meltdown***

A covert cyber breach that disables safety systems in nuclear facilities, leading to undetected reactor failure.

- ***Nuclear deterrence***

Strategy where countries prevent attacks by threatening nuclear retaliation.

- ***Nuclear proliferation***

The spread of nuclear weapons to more countries or groups.

- ***Mutually assured destruction***

Cold War doctrine where both sides avoid war because it would lead to total destruction.

- ***Launch-on-Warning***

A policy of launching nuclear weapons upon detecting an incoming attack.

- ***Supply chain compromise***

The insertion of vulnerabilities into hardware or software before installation in nuclear facilities.

III. Background Information

To gain a better grip on the gravity of the situation humanity finds itself in, one must understand the timeline of events starting from the beginning of the atomic age and following all grave tragedies, technological advances as well as the offensive and defensive undertakings of nuclear powers.

The atomic bombing of Hiroshima and Nagasaki

In 1945, the United States dealt the final blow to the Japanese Empire and ended the Second World War by deploying two nuclear bombs, named Fat Man and Little Boy, on Nagasaki and Hiroshima, respectively. The immediate death toll reached 120.000 and grew by up to 90.000 by the end of the year due to extreme burns and radiation sickness. The two incidents represent the only use of atomic bombs in a war.

The Cold War and the Nuclear Arms Race

During the Cold War, the Soviet Union and the United States invested vast amounts of money in the development of deadlier weaponry. In the realm of nuclear weapons, this meant the development of more powerful bombs, faster delivery methods and more sophisticated

detection systems. That being said, faulty technology and gear have led to incidents during which disaster was narrowly avoided thanks to last-minute human intervention or pure luck: the Cuban Missile Crisis, the 1983 Soviet Launch-on-Warning false alarm, or the 1979 NORAD computer error.

Chernobyl Disaster

On April 26th 1986, the global community witnessed the damage that a simple mismanagement of a nuclear power station can cause. The fateful late-night error alongside authorities' lack of preparedness led to catastrophic loss of human life and environmental damage. As such, the lethal capacity of civilian nuclear generators was proven.

David-Besse Slammer Worm

The Slammer worm infiltrated the information and communications systems at the Davis-Besse Nuclear Power Station in the United States through the infected device of an IT consultant. Although an accident, it showed how malware could reach critical energy infrastructure even when industrial networks were thought isolated.²

Stuxnet

Widely considered the first cyberweapon, Stuxnet was deployed by the United States and Israel in 2007 at Iran's uranium enrichment facility in Natanz, with the explicit goal of sabotaging operations. It operated for three years and did substantial damage to the facility's centrifuges.

The attack upon Poland's NCBJ

Following the war in Iran, Poland has reported that an alleged group of Iranian hackers³ attempted to infiltrate the country's National Centre for Nuclear Research (NCBJ), thus potentially breaching the country's only reactor. Despite the malicious group's failure, the attack has reignited panic with regards to cyber-nuclear threats.

² <https://uatom.org/en/2024/07/30/cybersecurity-in-nuclear-industry-protecting-digital-technologies-and-information.html>

³ <https://www.securityweek.com/hack-attempt-reported-at-polands-nuclear-research-center/>

IV. Major Countries and Organizations Involved

United States of America

The United States is the first state to develop and use nuclear weapons. In 1945, the United States dropped atomic bombs on the cities of Hiroshima and Nagasaki – the beginning of the nuclear age. During the Cold War, the United States was involved in a nuclear arms race with the Soviet Union, developing increasingly advanced launch and detection systems. In addition, the United States played a key role in the field of nuclear cybersecurity through the Stuxnet attack, carried out jointly with Israel against the Iranian nuclear programme. And even to this day, the United States is still one of the world's major holders of nuclear arsenals.

Islamic Republic of Iran

Iran began its nuclear programme in the 1950s, initially for energy and research purposes. Over time, its facilities have raised international concerns regarding the possibility of further developing nuclear weapons. In 2002, undeclared nuclear facilities were discovered at Natanz and Arak, leading to international negotiations and sanctions. In 2015, Iran signed The Joint Comprehensive Plan of Action⁴ in order to limit its nuclear activities. After the United States withdrawal from the deal in 2018, Iran resumed uranium enrichment at higher levels. In 2010, the Stuxnet cyberattack damaged centrifuges at Natanz and showed that nuclear infrastructure can be compromised without any actual military attack.

State of Israel

Between the 1950s and 1960s, Israel began its nuclear program in secret and built the “Dimona” complex⁵, where nuclear weapons were developed. Israel participated in the Stuxnet cyberattack, along with the United States, in order to sabotage Iranian nuclear facilities. Other cyber programs, such as “Flame”, were used to collect information from Iranian networks. In addition, Israel has carried out military attacks on Iranian facilities and personnel related to the nuclear programme, thus increasing tensions in the region.

⁴<https://www.ebsco.com/research-starters/diplomacy-and-international-relations/joint-comprehensive-plan-action-iran>

⁵<https://mondoweiss.net/2026/04/a-brief-history-of-the-israeli-nuclear-program-the-open-secret-at-the-heart-of-the-iran-war/>

Russian Federation

The Russian Federation, successor of the Soviet Union, owns one of the world's largest nuclear arsenals. During the Cold War, multiple incidents occurred – such as the 1983 false alarm⁶, when a US nuclear attack was mistakenly detected and an officer decided not to send the alert, thus avoiding war. In 1979, a miscalculation by NORAD computers⁷ revealed a US nuclear invasion, which was prevented in time by operators. After the collapse of the Union of Soviet Socialist Republics, Russia continued to modernise its ballistic missiles and nuclear submarines and was also found to be involved in several cyberattacks on foreign infrastructure.

People's Republic of China

China began its nuclear programme in the 1950s, with the support of France and the Soviet Union. In 1964, the first nuclear test was launched, becoming the fifth nuclear power in the world. From that point, China has expanded its arsenal and developed intercontinental ballistic missiles and nuclear submarines, combining conventional technology with the digitalisation of launching systems.

Democratic People's Republic of Korea

North Korea started its nuclear programme in the 1980s, initially for research and energy purposes only. Despite its initial plans, in 2006, the first nuclear test was recognised, following more tests of intercontinental ballistic missiles. North Korea is known for the cyberattacks on other states, including the theft of sensible information regarding technology and energy.

Republic of India

India began its nuclear programme in the 1940s-1950s for energy and research but did the first nuclear test in 1974, considered “peaceful”. Until 1998, India had already done multiple military nuclear tests, becoming a major worldwide nuclear power. Their ballistic missiles

⁶ <https://www.britannica.com/biography/Stanslav-Petrov>

⁷ <https://nsarchive.gwu.edu/briefing-book/nuclear-vault/2020-03-16/false-warnings-soviet-missile-attacks-during-1979-80-led-alert-actions-us-strategic-forces>

started being developed for national safety reasons, especially in the context of tensioned relations with Pakistan and rapprochement with China.

Islamic Republic of Pakistan

Pakistan began developing its own nuclear program in the 1970s in response to India's nuclear program. In 1998, it conducted nuclear tests and strengthened its deterrence doctrine, possessing strategic weapons and long-range missiles. Pakistan faces internal security challenges related to extremists and terrorism, which makes protecting its nuclear arsenal a constant priority. The state is modernising its infrastructure and delivery systems and collaborating with allies on defensive technologies. Rivalry with India remains the main factor influencing Pakistan's nuclear policy.

Ukraine

Upon independence, Ukraine inherited part of the Soviet Union's nuclear arsenal, which was the third largest in the world at the time. However, following international negotiations and non-proliferation agreements, Ukraine decided to give up its nuclear weapons and transfer them to Russia by 1996, in exchange for security guarantees through the Budapest Memorandum⁸. This decision was intended to reduce nuclear risk and integrate Ukraine into the international community. Even though Ukraine has given up its weapons, the state remains responsible for protecting its nuclear power plants and energy infrastructure, especially after the Chernobyl disaster and recent tensions with Russia. Military conflicts and cyber threats have only shown how vulnerable nuclear infrastructure can be and how important its surveillance and security are.



Image from the Chernobyl Nuclear Disaster

⁸ <https://www.armscontrol.org/factsheets/ukraine-nuclear-weapons-and-security-assurances-glance>

V. Timeline of Events

Date	Description of event
6 August 1945	The United States dropped the atomic bomb on Hiroshima, Japan.
9 August 1945	The United States dropped the second atomic bomb on Nagasaki, Japan.
16-28 October 1962	Tensions between the United States and the Soviet Union reached their peak. The USSR placed nuclear missiles in Cuba, and the US imposed a naval blockade.
23 June 1979	NORAD systems in the United States falsely indicated a Soviet nuclear attack.
26 September 1983	False nuclear alarm in the Soviet Union.
26 April 1986	A human error and lack of preparation caused the explosion of Reactor 4 at the Chernobyl nuclear power plant in Ukraine.
1996	Ukraine transfers its nuclear arsenal to Russia.
17 June 2007	The United States and Israel deployed the Stuxnet malware against Iran's uranium enrichment facility in Natanz. ⁹
January 2003	The Slammer worm infiltrated the communication and control systems of the David-Besse nuclear plant in the United States.
2010	Multiple cyberattacks ¹⁰ targeted Iranian nuclear facilities for sabotage and espionage.
2026	Attempted infiltration of Poland's NCBJ by Iranian hackers.

⁹ <https://en.wikipedia.org/wiki/Stuxnet>

¹⁰ https://en.wikipedia.org/wiki/Vulnerability_of_nuclear_facilities_to_attack

VI. Relevant UN Treaties and Events

- **United Nations Security Council Resolution 1373 (2001)**
- **United Nations Security Council Resolution 1540 (2004)**
- **United Nations Security Council Resolution 1887 (2009)**
- **United Nations Security Council Resolution 1977 (2011)**
- **United Nations Security Council Resolution 2325 (2016)**
- **Treaty on the Non-Proliferation of Nuclear Weapons (1968)**

The treaty establishes a framework for preventing the spread of nuclear weapons, promoting the peaceful uses of nuclear energy, and encouraging disarmament. All signatory states undertake not to transfer nuclear weapons to non-nuclear states and to negotiate in good faith for disarmament.¹¹

- **Convention on the Physical Protection of Nuclear Material (1980)**

Obliges states to protect nuclear material in domestic use, storage and transport.¹²

- **Comprehensive Nuclear-Test-Ban Treaty (1996)**

Bans all nuclear explosions, military or civilian. Although it has not yet entered into force, it establishes a global standard against nuclear testing and includes verification mechanisms to detect violations.¹³

- **International Convention for the Suppression of the Financing of Terrorism**

The convention aims to prevent and combat the financing of terrorism, emphasising international cooperation in addressing the financial support for terrorist activities.

¹¹<https://disarmament.unoda.org/en/our-work/weapons-mass-destruction/nuclear-weapons/treaty-non-proliferation-nuclear-weapons>

¹²<https://www.iaea.org/publications/documents/conventions/convention-physical-protection-nuclear-material-and-its-amendment>

¹³<https://www.ctbto.org/our-mission/the-treaty>

VII. Previous Attempts to Solve the Issues

- **International Atomic Energy Agency (IAEA)**

The IAEA was created on the 29th of July 1957, in order to monitor the civil nuclear programs of signatory states for the prevention of their use for military purposes. Periodic inspections and controls were designed to detect any deviation from the civilian purposes of nuclear energy. Through these checks, the IAEA provided international transparency, reduced the risks of nuclear proliferation, and created a framework in which cooperation between states was possible.¹⁴

- **Budapest Memorandum on Security Assurances**

Through the Budapest Memorandum, created on the 5th of December 1994, Ukraine agreed to give up its nuclear weapons in exchange for security guarantees from Russia, the United States, and the United Kingdom. This action was an attempt to prevent nuclear proliferation by offering a non-military alternative to the possession of nuclear weapons.¹⁵

- **Joint Comprehensive Plan of Action (JCPOA)**

The JCPOA, created on the 14th of July 2015 and signed in Budapest, Hungary, was an international cooperation agreement between Iran and the P5+1 (US, UK, France, Russia, China + Germany) with the purpose of limiting Iran's nuclear program in exchange for the lifting of economic sanctions. Iran committed to reducing the number of centrifuges, limiting the level of uranium enrichment, and allowing regular IAEA inspections. The agreement was a direct attempt to prevent nuclear proliferation and showed that diplomacy and international monitoring can be effective without resorting to military force.

VIII. Possible Solutions

¹⁴ <https://www.iaea.org>

¹⁵ <https://www.hks.harvard.edu/publications/budapest-memorandum-25-between-past-and-future>

Addressing the issue of nuclear threats and cybersecurity in nuclear facilities is complex and it requires a comprehensive and coordinated approach that involves not only national governments but also international organisations, nuclear regulatory bodies, and cybersecurity experts. Strengthening physical security, enhancing cyber defence, and implementing risk-reduction programmes are essential components of any effective strategy to tackle nuclear and cyber risks.

- **Physical Security Improvements**

Enhancing the physical security of nuclear facilities is a crucial factor in order to help reduce the number of unauthorised accesses or sabotage attempts. Seeing as poorly protected installations are one of the main facilitators of nuclear and cyber threats, it is imperative that nations take action towards strengthening their security measures. This involves implementing coordinated and comprehensive protection strategies, which require close collaboration among competent authorities at the facility locations. Strengthening inter-agency cooperation and international partnerships can further amplify these efforts, ensuring that protection measures are seamless and effective.¹⁶

- **Cybersecurity Collaboration**

Effective cybersecurity measures and information sharing among nuclear states are vital for preventing cyberattacks and compromising of nuclear infrastructure. By establishing strong communication channels and collaborative frameworks, countries can share critical information regarding vulnerabilities, malware threats, and cyberattack tactics. By sending trained cybersecurity professionals to high-risk facilities, we can much more safely take control of the issue and tackle it firsthand. This collaborative approach enables timely interventions and facilitates collaboration towards the common goal of global nuclear safety.¹⁷

- **Risk Reduction Programs**

Risk-reduction initiatives play a significant role in decreasing the availability of materials and vulnerabilities that could be exploited in nuclear or cyber-attacks. By ensuring peaceful communication between key actors and regular monitoring of nuclear infrastructure, we can reach a consensus that gradually leads to a secure and well-protected environment.

¹⁶ <https://world-nuclear.org/information-library/safety-and-security/security/security-of-nuclear-facilities-and-material>

¹⁷ <https://www.nrc.gov/security/cybersecurity>

- **Cessation of Geopolitical Conflicts**

Conflicts between nuclear-armed states or between states with advanced nuclear programmes only serve to worsen the global situation. Rivalries, such as India-Pakistan or tensions between Russia and Ukraine, increase the risks of nuclear or cyber incidents. To pacify nuclear threats, countries must first make peace with each other and coordinate security efforts internationally.

Questions delegates should consider during Research

- o How is your country's national security affected by nuclear threats and cyber vulnerabilities in nuclear infrastructure?
- o Is your country actively developing or modernising nuclear weapons or related delivery systems?
- o What measures has your country implemented to protect nuclear facilities against physical and cyber-attacks?
- o Is your country involved in international or regional security alliances or cooperative initiatives that address nuclear and cyber threats?
- o What international agreements, frameworks or resolutions on nuclear security and non-proliferation is your country a signatory to?
- o What are the root causes of nuclear proliferation and cybersecurity risks in the nuclear sector? How can they be addressed?

IX. Sources

- *Introduction:*

<https://www.dw.com/en/biden-putin-summit-experts-warn-of-cyber-nuclear-nightmare/a-57906147>

- *Background Information:*

<https://uatom.org/en/2024/07/30/cybersecurity-in-nuclear-industry-protecting-digital-technologies-and-information.html>

- *Major Countries Involved:*

<https://www.securityweek.com/hack-attempt-reported-at-polands-nuclear-research-center/>

<https://www.ebsco.com/research-starters/diplomacy-and-international-relations/joint-comprehensive-plan-action-iran>

<https://mondoweiss.net/2026/04/a-brief-history-of-the-israeli-nuclear-program-the-open-secret-at-the-heart-of-the-iran-war/>

<https://www.britannica.com/biography/Stanslav-Petrov>

<https://nsarchive.gwu.edu/briefing-book/nuclear-vault/2020-03-16/false-warnings-soviet-missile-attacks-during-1979-80-led-alert-actions-us-strategic-forces>

<https://www.armscontrol.org/factsheets/ukraine-nuclear-weapons-and-security-assurances-glance>

- *Timeline of Events:*

<https://en.wikipedia.org/wiki/Stuxnet>

https://en.wikipedia.org/wiki/Vulnerability_of_nuclear_facilities_to_attack

- *Relevant UN Treaties and Events:*

<https://disarmament.unoda.org/en/our-work/weapons-mass-destruction/nuclear-weapons/treaty-non-proliferation-nuclear-weapons>

<https://www.iaea.org/publications/documents/conventions/convention-physical-protection-nuclear-material-and-its-amendment>

<https://www.ctbto.org/our-mission/the-treaty>

- *Previous Attempts to Solve the Issue:*

<https://www.iaea.org>

<https://www.hks.harvard.edu/publications/budapest-memorandum-25-between-past-and-future>

- *Possible Solutions:*

<https://world-nuclear.org/information-library/safety-and-security/security/security-of-nuclear-facilities-and-material>

<https://www.nrc.gov/security/cybersecurity>

- *Further Reading and Useful Links:*

<https://carnegieendowment.org/research/2025/09/preventing-nuclear-anarchy-nuclear-proliferation-and-american-security>

<https://www.rusi.org/explore-our-research/publications/occasional-papers/cyber-threats-and-nuclear-weapons-new-questions-command-and-control-security-and-strategy>

<https://www.weforum.org/stories/2018/02/nuclear-weapons-and-cybersecurity-threats-vulnerabilities-and-consequences/>

<https://www.nti.org/analysis/articles/cyber/>

<https://www.armscontrol.org/factsheets/nuclear-weapons-who-has-what-glance>

